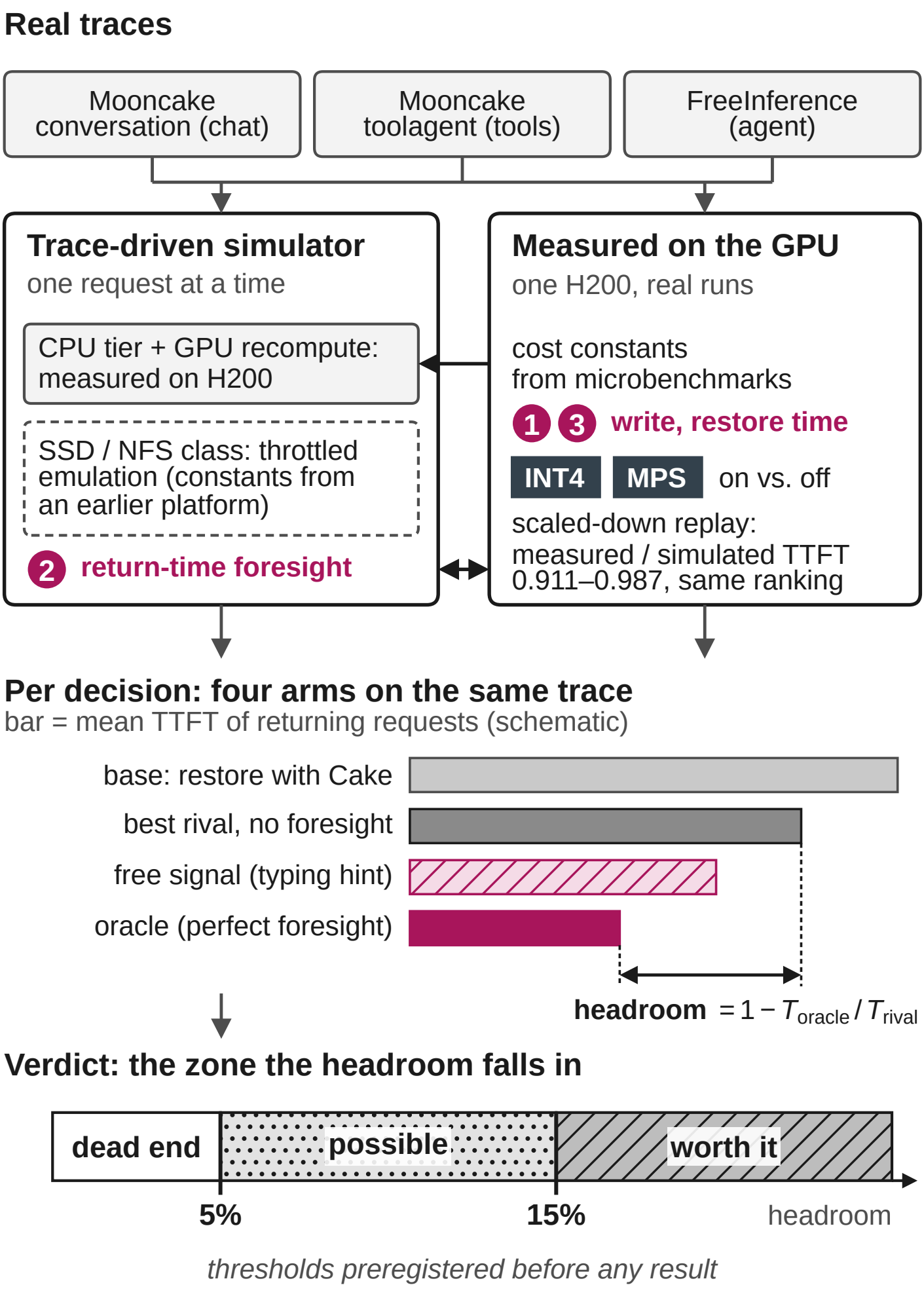

Real traces
Mooncake conversation (chat)
Mooncake toolagent (tools)
FreeInference (agent)
Trace-driven simulator
one request at a time
CPU tier + GPU recompute: measured on H200
SSD / NFS class: throttled emulation (constants from an earlier platform)
2 return-time foresight
Measured on the GPU
one H200, real runs
cost constants from microbenchmarks
1 3 write, restore time
INT4 MPS on vs. off
scaled-down replay: measured / simulated TTFT 0.911–0.987, same ranking
Per decision: four arms on the same trace
bar = mean TTFT of returning requests (schematic)
base: restore with Cake
best rival, no foresight
free signal (typing hint)
oracle (perfect foresight)
headroom = 1 − T oracle / T rival
Verdict: the zone the headroom falls in
dead end
possible
worth it
5%
15%
headroom
thresholds preregistered before any result